

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2021

Math 1113

(Differential and Integral Calculus)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define domain and range of a function. Find the domain and range of $y = \frac{1}{\sqrt{|x|-x}}$ 10
- 1(b). Show that the function 15
- $$f(x) = x \cos\left(\frac{1}{x}\right), \quad x \neq 0$$
- $$= 0, \quad x = 0$$
- is continuous at $x = 0$, but has no derivative there.
- 1(c). If $y = e^{ax} \cos(bx + c)$, then find the n th derivative of y . 10
- 2(a). Define limit of a function. Evaluate 10
- $$\lim_{x \rightarrow 1} \left[\frac{x}{x-1} - \frac{1}{\ln x} \right]$$
- 2(b). If $y = \cos(m \sin^{-1} x)$, prove that 15
- (i) $(1 - x^2)y_2 - xy_1 + m^2y = 0$
- (ii) $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} + (m^2 - n^2)y_n = 0$
- 2(c). State Rolle's theorem. Verify Rolle's theorem for the function $f(x) = 1 - x^{2/3}$ over the interval $[-1, 1]$. 10
- 3(a). If $u = \cos^{-1} \frac{x+y}{\sqrt{x} + \sqrt{y}}$, then find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$. 12
- 3(b). Find the radius of curvature at the point (r, θ) on the cardioide $r = a(1 - \cos \theta)$, and show that it varies as $\sqrt{r}$. 12
- 3(c). Find the value of $\sin 31^\circ$ by using Taylor's theorem. 11
- 4(a). Discuss the maximum and minimum of the function $f(x) = 5x^6 - 18x^5 + 15x^4 - 10$. 12
- 4(b). Expand $\log_e x$ in power of $(x - 1)$ and hence evaluate $\log_e(1.1)$ correct to 4 decimal places. 12
- 4(c). Find $\frac{dy}{dx}$, where $x^y + (\sin x)^{\cos x} = x + y$. 11

SECTION – B

5. Integrate the followings:

(a). $\int \frac{dx}{(x^2 + 4)\sqrt{x^2 + 9}}$ 12

(b). $\int \sin^{-1}\left(\sqrt{\frac{x}{x+a}}\right) dx$ 11

(c). $\int \frac{dx}{1 + 3e^x + 2e^{2x}}$ 12

6. Evaluate the followings:

(a). $\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x} dx$ 12

(b). $\int_0^{\pi/4} \frac{x}{1 + \cos 2x + \sin 2x} dx$ 12

(c). $\lim_{n \rightarrow \infty} \left[\frac{1^2}{n^3 + 1^3} + \frac{2^2}{n^3 + 2^3} + \dots + \frac{n^2}{2n^3} \right]$ 11

7(a). Find the reduction formula for $\int \sec^n x dx$ and hence find $\int \sec^3 x dx$ 13

7(b). Find the reduction formula for $\int \frac{dx}{\sin^m x \cos^n x}$ 11

7(c). Evaluate $\int_0^1 x^4 (1 - x^2)^{3/2} dx$ by using the rule of Beta function. 11

8(a). Find the length of the arc of the parabola $y^2 = 4ax$ cut off by the line $3y = 8x$. 12

8(b). Find the volume of the solid produced by the revolution of the loop of the curve $y^2(a + x) = x^2(a - x)$ about the x-axis. 11

8(c). Find the area of a loop of the curve $a^2 y^2 = x^2(a^2 - x^2)$ 12

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2021

Ch 1113

(Chemistry I)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|--|----|
| 1(a). | Define reversible and irreversible reactions. "Equilibrium can be initiate from either side of a chemical reaction" – explain with diagram. | 10 |
| 1(b). | What is standard free energy? Derive a relation between standard free energy change and equilibrium constant of a reaction at a given temperature. | 10 |
| 1(c). | Calculate the free energy change (ΔG^0) associate with chemical reaction for which the equilibrium constant at 298 K is 0.01. ($R = 8.314 \text{ J. mol}^{-1}\text{K}^{-1}$) | 07 |
| 1(d). | "Chemical equilibrium is a dynamic process" – explain. | 08 |
| | | |
| 2(a). | Mention the classification of batteries. Explain with diagram the reaction mechanism of the lithium ion battery at discharge state. | 10 |
| 2(b). | Draw a thermic diagram of proton exchange membrane fuel cell and explain it's varius components. | 10 |
| 2(c). | Calculate the EMF of the cell:
$\text{Zn} \text{Zn}^{2+}(0.001M) \parallel \text{Ag}^+(0.1M) \text{Ag}$
The standard potential of Ag/Ag^+ half cell is 0.080 V and $\text{Zn} \text{Zn}^{2+}$ is -0.76 V . | 07 |
| 2(d). | Explain why emf of an electrochemical cell can not be measured with the help of voltameter? | 08 |
| | | |
| 3(a). | Draw a thermic diagram of a nuclear reactor and explain its different components. | 10 |
| 3(b). | What are the differences between natural and artificial radio-activity? Describe artificial disiutegration of atoms with suitable examples. | 10 |
| 3(c). | How many α -particles per second would be emitted from $4 \times 10^{-12} \text{ g}$ of ^{210}Po ($t_{1/2} = 138 \text{ days}$)? | 07 |
| 3(d). | What is radioactivity? Describe the properties of the radiation obtained from radioactive materials. | 08 |

4(a).	Define over potential. Explain electrical double layer based on Helmholtz model.	10
4(b).	Draw a cyclic voltogram and indicate its different sections.	10
4(c).	State the Kohlrausch's law of independent migration of ions. How can you determine Λ_0 of weak electrolytes using this law?	10
4(d).	What is electrochemical series?	05

SECTION – B

5(a).	Why all macromolecules are not polymer?	08
5(b).	n-butyl bromide reacts with lithium in presence of ether the product which is formed here is used to initiate a polymerization reaction. Discuss the mechanism of this polymerization reaction where styrene acts as a monomer and water acts as a chain terminating agent.	12
5(c).	How would you differentiate thermos-setting polymer from thermoplastic polymer?	10
5(d).	The degree of polymerization of the polyvinyl acetate is 1000. Calculate the molecular weight of polyvinyl acetate.	5
6(a).	Define with example of the following terms:	10
	(i) Base unit	
	(ii) Repeating unit	
	(iii) End group	
	(iv) Network polymer	
	(v) Inorganic polymer	
6(b).	How will you prepare Nylon-66 and Dacron?	10
6(c).	What is number average molecular weight and weight average molecular weight? When $\bar{M}_w$ and $\bar{M}_n$ are to be equal?	08
6(d).	Mention some examples of conducting polymer. Why conducting polymer has advantages over traditional polymer.	07
7(a).	What are the harmful effects of oxides of Sulphur and Nitrogen in atmosphere?	15
7(b).	What is arsenicosis? Mention the mechanism basis of arsenicosis.	10
7(c).	What are the types of water quality parameters? Mention the indicators that help to understand the water quality.	10
8(a).	Though Sulphur increases the calorific value of coal, the presence of Sulphur in coal is undesirable. Why?	07
8(b).	How will you synthesize gasoline?	12
8(c).	What is petroleum? What are the advantages of catalytic cracking over thermal cracking?	08
8(d).	What are meant by water gas and producer gas? Mention some uses of them.	08

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2021

Ph 1113

(Physics)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

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|-------|--|----|
| 1(a). | Explain oscillatory behaviour of simple harmonic motion. | 12 |
| 1(b). | Show that for a particle executing simple harmonic motion, its velocity at any instant is $\omega\sqrt{a^2 - y^2}$. | 10 |
| 1(c). | Derive an expression of time period for simple pendulum. | 13 |
| 2(a). | What is Lissayous' figure? Write down its use. | 07 |
| 2(b). | Claculate the resultant of two simple harmonic motions of equal time period when they act at right angles to one another and when the phase difference is zero and $\pi/2$. | 18 |
| 2(c). | Draw Lissayous' figure using equations $x = a \sin 2\omega t$ and $y = 2a \sin \omega t$. | 10 |
| 3(a). | Derive differential equations of undamped, damped and forced vibrations. | 12 |
| 3(b). | Write down different forms of equation of travelling wave, hence derive differential equation of wave motion. | 09 |
| 3(c). | Discuss logarithmic decrement and quality factor of damped harmonic oscillation. | 10 |
| 3(d). | What are stationary waves? | 04 |
| 4(a). | Why c_p is greater than c_v ? | 05 |
| 4(b). | Obtain an expression for electrical conductivity of a metal on the basis of free electron theory. | 10 |
| 4(c). | Calculate the number of atoms per unit cell of body-centered cube. | 10 |
| 4(d). | Calculate the dimentions of unit cell of body centered cube. | 10 |

SECTION – B

- | | | |
|-------|---|----|
| 5(a). | Discuss the details observations of photoelectric effect? | 10 |
| 5(b). | In what sense, wave theory is inadequate to explain photoelectric effect? Discuss Einstein's explanation regarding the photoelectric effect. | 15 |
| 5(c). | X-ray of wavelength 0.55×10^{-10} m are scattered by free electrons in a block of carbon through 90° . Find the energy of (i) incident photons, (ii) scattered photons, (iii) recoil electrons and (iv) momentum of recoil electrons. | 10 |
| | | |
| 6(a). | What is de Broglie wave? Discuss wave function of de Broglie wave. | 10 |
| 6(b). | What is the difference between de Broglie wave velocity and group velocity? Show that the group velocity is equal to the velocity of the particle. | 15 |
| 6(c). | What is the de Broglie wavelength of an electron whose speed is 9×10^7 m/sec? | 10 |
| | | |
| 7(a). | On the basis of Bohr's atom model, derive an expression for the spectral series of Hydrogen atom. | 15 |
| 7(b). | Write down all quantum number related to Vector Atom model. | 10 |
| 7(c). | Calculate the age of the earth with the help of radio-active decay. | 10 |
| | | |
| 8(a). | Define fission and fusion with example hence explain binding energy curve with fission and fusion. | 10 |
| 8(b). | What is the difference between natural and artificial radioactivity?

Prove that $N = N_0 e^{-\lambda t}$, where the symbols have their usual meanings.

Hence show that $T_{1/2} = \frac{0.693}{\lambda}$. | 15 |
| 8(c). | The half-life of ${}_{92}\text{U}^{238}$ against alpha decay is 4.5×10^9 years. How much disintegration per second occurs in 1 kg of Uranium? | 10 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2021

Hum 1113

(Sociology and Behavioral Science)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|---|----|
| 1(a). | What is meant by “Sociology”? | 05 |
| 1(b). | Discuss shortly the origin and scope of sociology? | 15 |
| 1(c). | Explain the importance of sociology for the students of Energy Science and Engineering. | 15 |
| 2(a). | What is social stratification? | 05 |
| 2(b). | Mention the characteristics of society and community. | 15 |
| 2(c). | Explain the important functions of family. | 15 |
| 3(a). | What is meant by socialization? | 05 |
| 3(b). | Discuss the society and culture of Bangladesh. | 15 |
| 3(c). | Discuss the causes of crime and juvenile delinquency. | 15 |
| 4(a). | What is industrialization? | 05 |
| 4(b). | Discuss the impact of industrialization on our society and environment. | 15 |
| 4(c). | What is social structure? Give the concept of Karl Marx’s social structure. | 15 |

SECTION – B

- | | | |
|-------|--|----|
| 5(a). | What is organizational behavior? Why organizational behavior is needed? | 10 |
| 5(b). | What is training? Discuss the different steps of training process. | 15 |
| 5(c). | Discuss the human behavior. | 10 |
| 6(a). | What is meant by employee relation? Explain the way to improve employee relation in any organization. | 15 |
| 6(b). | What is unemployment? Discuss the social impacts of unemployment. | 10 |
| 6(c). | Describe the dimensions of employee relation. | 10 |
| 7(a). | What is leadership? Discuss the traits of effective leaders. | 15 |
| 7(b). | What is leadership style? Discuss the different types of leadership style. Which of them do you like most and why? | 20 |
| 8(a). | What is stress? What are the causes of stress of stress in workplace? | 10 |
| 8(b). | What is meant by counseling? Explain role of counseling in reducing stress of employees. | 15 |
| 8(c). | What is motivation? Discuss the effectiveness of Alderfer’s ERG model in workplace. | 10 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2021

ESE 1101

(Fundamentals of Energy Resources)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- | | | |
|-------|---|----|
| 1(a). | Define energy and energy resources. How the natural resources are classified? Explain in brief. | 10 |
| 1(b). | What is meant by earth energy cycle? Explain the earth energy cycle with necessary sketch. | 10 |
| 1(c). | Distinguish between the followings:
i). Primary and secondary resources of energy.
ii). Commercial and non commercial energy. | 10 |
| 1(d). | Describe the adverse effect of non-renewable energy sources. | 05 |
| | | |
| 2(a). | Write a short note on history of energy uses. Write different forms of energy with application. | 12 |
| 2(b). | Briefly discuss about the current status of non-conventional energy sources of Bangladesh. | 10 |
| 2(c). | What are the different types of coal? “Peat is not actually coal ,but rather the precursor to coal” Explain. | 08 |
| 2(d). | Describe the disadvantages of using coal to produce electricity. | 05 |
| | | |
| 3(a). | What is crude oil? Explain different physical properties of petroleum oil. | 10 |
| 3(b). | Illustrate the origin and formation of petroleum along with organic and inorganic theories. | 15 |
| 3(c). | What are oil shale and tar sands? | 05 |
| 3(d). | What are the different types of crude oil based on hydrocarbon? Mention the uses of each of them. | 05 |
| | | |
| 4(a). | What is meant by natural gas exploration? Explain in brief. | 08 |
| 4(b). | What are the different form of natural gas? Mention their uses. Discuss about precaution of uses of natural gas. | 09 |
| 4(c). | With a schematic diagram, explain the working principle of a nuclear power plant. | 12 |
| 4(d). | Discuss about the challenges of use of nuclear energy. | 06 |

SECTION – B

- | | | |
|-------|--|----|
| 5(a). | Explain the process of solar energy formation. Also explain the attenuation process of solar radiation from sun surface to earth surface. | 10 |
| 5(b). | What are meant by i) beam radiation, ii) diffuse radiation, iii) global radiation.
How to measure the solar radiation on earth surface? | 09 |
| 5(c). | With a neat sketch describe how portable water is obtained from salt water by using solar energy. | 08 |
| 5(d). | Why dopping is necessary? Explain with a neat sketch how solar energy can be converted into direct electricity. | 08 |
| | | |
| 6(a). | Define biomass, bioenergy and bio-fuel. Why biomass conversion do is necessary? Explain. | 09 |
| 6(b). | Classify biomass conversion system. With a neat sketch, explain the biomass gasification process. | 10 |
| 6(c). | Briefly explain the fermentation process for the production of bioethanol. | 08 |
| 6(d). | What are the different sources of biomass? Explain the parameters to be controlled to achieve the biomass production efficiently. | 08 |
| | | |
| 7(a). | Explain the term “pumped storage hydro power plat”. With a neat sketch, describe how hydropower plant produces electricity from water. | 10 |
| 7(b). | What are the different types of turbines used in hydropower plant? | 05 |
| 7(c). | What are the differences between tidal energy and hydropower? How wave energy can be used to generate electricity? Explain. | 13 |
| 7(d). | How tidal energy is converted into power? | 07 |
| | | |
| 8(a). | How wind is formed? With neat sketch, explain HAWT and VAWT. | 10 |
| 8(b). | Do wind turbines impact on human health? Does wind need backup or storage? Explain | 05 |
| 8(c). | How to extract geothermal energy? Explain. Also mentioned the application of geothermal energy. | 10 |
| 8(d). | Explain the term OTEC. With neat sketch describe open cycle OTEC system. | 10 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 1st Term Examination, 2021

ESE 4007

(Smart Grid Technology)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is a smart grid? Compare it with the conventional grid. 10
- 1(b). Explain the roles of stakeholders in smart grids. 10
- 1(c). Provide an overview of the following technology required for smart grids. 15
(i) Information and communication technologies.
(ii) Sensing, measurement, control and automation technologies.
(iii) Power electronics and energy storage.
- 2(a). What is a microgrid? Describe the operation of distribution network active management scheme with necessary block diagram. 12
- 2(b). Define HAN, NAN and WAN. Write down the application areas of the following communication technologies in smart grids: 11
Bluetooth, Wi-Fi, ZigBee, PLC and WiMAX.
- 2(c). Write down the necessity of GIS and google mapping tools in smart grids. 12
- 3(a). Compare the conventional and smart meter. 10
- 3(b). What is meant by demand side management? Describe how the following techniques modify the load consumption patterns. 15
(i) Load shifting, (ii) Valley filling, and (iii) Peak clipping.
- 3(c). Consider the circuit shown in figure 3(C1). The 33/11 kV transformer has an on-load tap changer which maintains the load voltage at 11 kV. Calculate the percentage reduction in energy loss in the 33 kV line if load shifting shown in figure 3(C2) is managed. Ignore the 33/11 kV transformer losses. 10

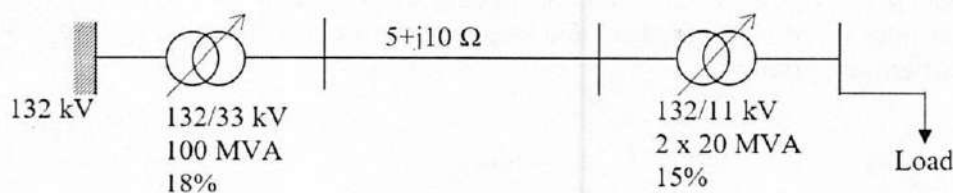


Figure 3(C1)

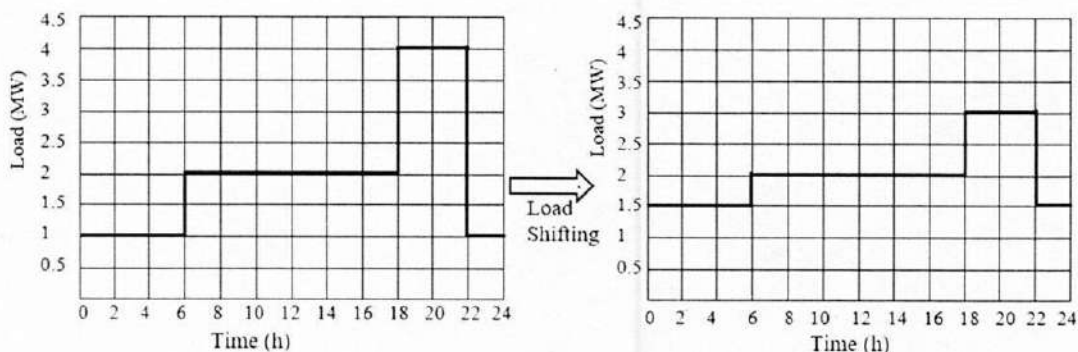


Figure 3(C2)

- | | | |
|-------|---|----|
| 4(a). | Describe the application of multi-agent system (MAS) in smart grid with an example. What are the advantages of the multi-agent approach? | 15 |
| 4(b). | Describe the weaknesses of the conventional load flow methods. What new features of the load flow make it suitable for smart grid applications? | 10 |
| 4(c). | Write a short note on static security assessment (SSA) of smart power systems. | 10 |

SECTION – B

- | | | |
|-------|---|----|
| 5(a). | What is stability in smart grid? Why is power quality management in smart grid so important? | 06 |
| 5(b). | What is meant by Load Frequency Control (LFC) and Automatic Voltage Regulator (AVR)? Also explain their role in smart grid with schematic diagram. | 12 |
| 5(c). | What is the necessity of reactive power control in power systems? Describe the voltage/VAr control technique in smart grid networks. | 09 |
| 5(d). | “The gradual increase of the generator power output is possible until the maximum electrical power is transferred”. Justify the statement. | 08 |
| 6(a). | Write short notes on (i) voltage stability, and (ii) voltage collapse. | 12 |
| 6(b). | Give an overview of the methods that are used for the assessment of angle stability of power systems. | 10 |
| 6(c). | Define Expert Systems (ES) and write the features of ES. Illustrate the fundamental components of an expert system. | 13 |
| 7(a). | What is intelligent grid operation? Describe the necessity of bulk power system automation of smart grid at transmission level. | 15 |
| 7(b). | How smart grid improve environmental sustainability and promotes greener future? | 10 |
| 7(c). | Describe the barriers of smart grid. | 10 |
| 8(a). | What are the benefits and challenges of interoperability? | 12 |
| 8(b). | Describe the impact of cyber security on the smart grid. | 12 |
| 8(c). | What are the techniques available for smart grid optimization? Describe computational challenges faced by researchers and engineers for design, planning and operations of smart power systems. | 13 |

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